CHAPTER 1

Introduction to Machine Learning

Machine learning is a branch of artificial intelligence that learns mathematical models from data and is broadly categorized into supervised, unsupervised, and reinforcement learning, all of which are supported by deep neural networks (Figure 1.1).

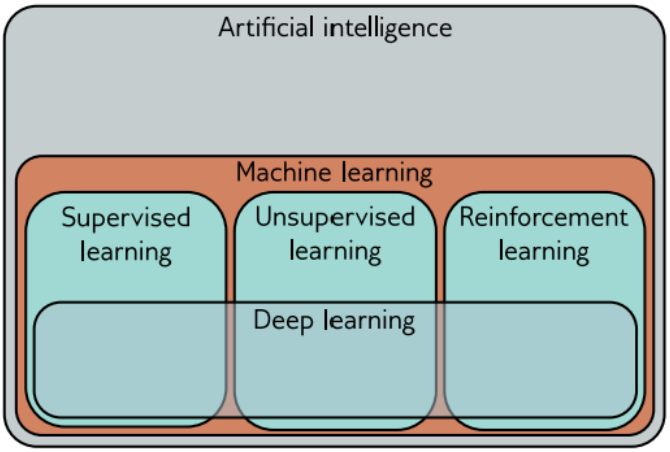

Figure 1.1 Overview of Machine Learning Paradigms

1.1. Supervised learning

Supervised learning models learn a mapping from input data to output predictions by training on labeled examples, where each input is paired with a known target. The objective is to infer a function that can accurately predict the correct output for new, unseen inputs.

1.1.1. Regression and classification problems

Figure 1.2 illustrates several regression and classification tasks in which real-world inputs (such as text, audio, or images) are first encoded as numerical vectors. These vectors serve as model inputs, which are then mapped to output vectors that are decoded into meaningful real-world predictions.

- The model shown in Figure 1.2a estimates a house's price from features like square footage and number of bedrooms. This is a regression task, as the output is a continuous value rather than a discrete category.
- The model in Figure 1.2b uses a molecule's chemical structure as input to predict its freezing and boiling points. This constitutes a multivariate regression problem because it outputs multiple continuous values.
- The model in Figure 1.2c takes a restaurant review as input and predicts whether it is positive or negative. This is a binary classification task, as the goal is to assign the input to one of two categories. The output vector represents the probabilities of the input belonging to each category.
- Figures 1.2d and 1.2e illustrate multiclass classification tasks, where the model assigns each input to one of $N > 2$ categories. In Figure 1.2d, the input is an audio file, and the model predicts its music genre. In Figure 1.2e, the input is an image, and the model predicts the object it contains. In both cases, the output is an N-dimensional vector representing the probabilities for each category.

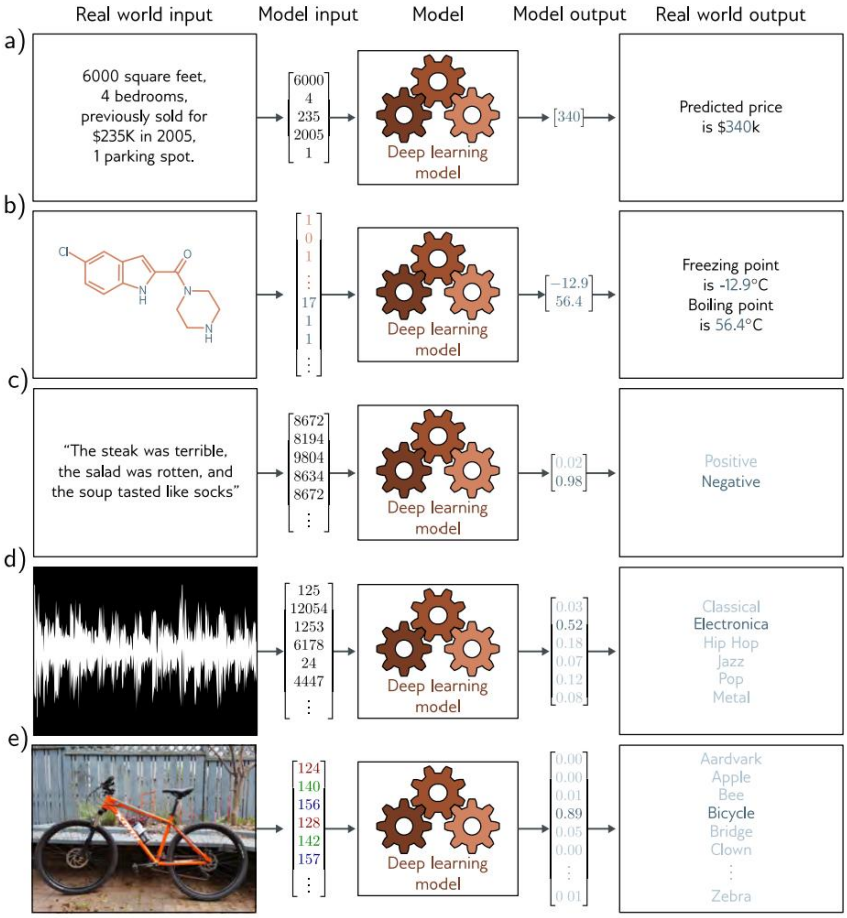

Figure 1.2 Regression and classification problems.

1.1.2. Inputs

The input data in Figure 1.2 comes in diverse forms:

- In the house pricing example, the input is a fixed-length vector representing property features. This is considered tabular data, as it lacks internal structure—reordering the input features and retraining the model would not affect its predictions.
- The input in the restaurant review example is text, which can vary in length depending on the number of words. The order of words is crucial— "my wife ate the chicken" conveys a completely different meaning than "the chicken ate my wife." Before feeding the text into the model, it must be converted into a numerical representation. In this case, a fixed vocabulary of 10,000 words is used, and each review is represented by concatenating the indices of its words.
- In the music classification example, the input may be a fixed-length vector, such as a 10-second audio clip, but it is extremely high-dimensional. Digital audio is typically sampled at 44.1 kHz with 16-bit resolution, meaning a ten-second clip contains 441,000 integers.
- In the image classification example, the input—formed by concatenating the RGB values of all pixels—is also very large. Additionally, it has spatial structure: pixels that are vertically or horizontally near each other are closely related, even if they are far apart in the flattened input vector.
- Finally, consider the input for the model predicting a molecule's freezing and boiling points. Molecules can have varying numbers of atoms arranged in different structures. Therefore, the model must process both the molecule's geometric structure and the types of atoms it contains.

1.1.3. Structured outputs

The structured output data in Figure 1.3 comes in diverse forms:

- Figure 1.3a illustrates a multivariate binary classification model for semantic segmentation, where each pixel in an input image is labeled to indicate whether it belongs to a cow or the background. Figure 1.3b shows a multivariate regression model that predicts the depth at every pixel from a street scene image. In both cases, the outputs are high-dimensional and structured, with the structure closely related to the input—for example, if one pixel is labeled as "cow," neighboring pixels with similar RGB values are likely to share the same label.
- Figures 2.3c–e depict models whose outputs have complex structures that are less directly tied to the inputs. In Figure 2.3c, the model transcribes an audio file into spoken words. Figure 2.3d shows a translation model that converts English text into French. Figure 2.3e illustrates a particularly challenging task: generating an image from descriptive text.

While these tasks can, in principle, be approached using standard supervised learning, they are more difficult for two main reasons. First, the outputs can be inherently ambiguous—there are multiple valid translations for a given English sentence and many possible images that match a single caption. Second, the outputs have rich internal structure; not all sequences of words form valid English or French sentences, and not all combinations of RGB values produce plausible images. This means that, beyond learning the mapping, the model must also respect the "grammar" or structural rules of the output.

Fortunately, this output "grammar" can often be learned without explicit labels. For instance, valid English sentence structures can be learned from the statistical patterns in a large text corpus. This idea naturally leads into the next section, which introduces unsupervised learning models.

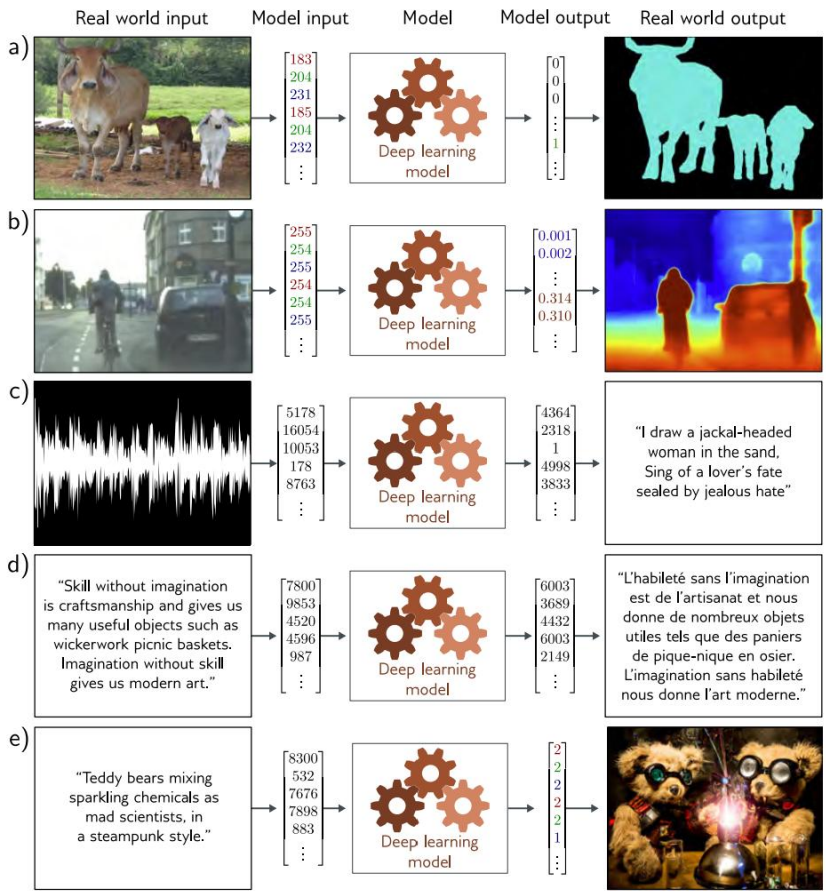

Figure 1.3 Supervised learning tasks with structured outputs.

1.2. Unsupervised learning

Unsupervised learning refers to building a model from input data without associated output labels, meaning there is no "supervision." Instead of learning a mapping from inputs to outputs, the goal is to uncover or describe the underlying structure of the data. Similar to supervised learning, the input data can vary widely—it may be discrete or continuous, low- or high-dimensional, and of fixed or variable length.

1.2.1. Generative models

Generative unsupervised models aim to create new data samples that are statistically similar to the training data. Some of these models explicitly model the probability distribution of the input data, allowing new examples to be generated by sampling from this distribution—for instance, Variational Autoencoders (VAEs).

Other models generate new examples without explicitly modeling the underlying data distribution. Examples include Generative Adversarial Networks (GANs), Diffusion Models (such as DALL·E 2 and Stable Diffusion), and generative Transformers (like GPT or BERT for masked token prediction).

State-of-the-art generative models can synthesize examples that are extremely plausible but distinct from the training examples. They have been particularly successful at generating images (figure 1.4) and text (figure 1.5).

Figure 1.4 Generative models for images. Left: two images were generated from a model trained on pictures of cats. These are not real cats, but samples from a probability model. Right: two images generated from a model trained on images of buildings.

The moon had risen by the time I reached the edge of the forest, and the light that filtered through the trees was silver and cold. I shivered, though I was not cold, and quickened my pace. I had never been so far from the village before, and I was not sure what to expect. I had been walking for hours, and I was tired and hungry. I had left in such a hurry that I had not thought to pack any food, and I had not thought to bring a weapon. I was unarmed and alone in a strange place, and I did not know what I was doing.

I had been walking for so long that I had lost all sense of time, and I had no idea how far I had come. I only knew that I had to keep going. I had to find her. I was getting close. I could feel it. She was nearby, and she was in trouble. I had to find her and help her, before it was too late.

Figure 1.5 Short story synthesized from a generative model of text data.

Generative models can also produce data while enforcing specific constraints on certain outputs, a process known as conditional generation. Examples include image inpainting (Figure 1.6) and text completion (Figure 1.7).

Figure 1.6 Inpainting. In the original image (left), the boy is obscured by metal cables. These undesirable regions (center) are removed and the generative model synthesizes a new image (right) under the constraint that the remaining pixels must stay the same.

I was a little nervous before my first lecture at the University of Bath. It seemed like there were hundreds of students and they looked intimidating. I stepped up to the lectern and was about to speak when something bizarre happened.

Suddenly, the room was filled with a deafening noise, like a giant roar. It was so loud that I couldn't hear anything else and I had to cover my ears. I could see the students looking around, confused and frightened. Then, as quickly as it had started, the noise stopped and the room was silent again.

I stood there for a few moments, trying to make sense of what had just happened. Then I realized that the students were all staring at me, waiting for me to say something. I tried to think of something witty or clever to say, but my mind was blank. So I just said, "Well, that was strange,' and then I started my lecture.

Figure 1.7 Conditional text synthesis. Given an initial body of text (in black), generative models of text can continue the string plausibly by synthesizing the "missing" remaining part of the string.

1.2.2. Latent variables

Some generative models—though not all—leverage the observation that real data often lie on a space of much lower dimensionality than the raw number of observed variables would suggest. For instance, the set of valid and meaningful English sentences is vastly smaller than the set of all possible word sequences formed at random. Likewise, real-world images constitute only a minute fraction of all images that could be produced by assigning random red, green, and blue (RGB) values to each pixel. This restriction arises because images are shaped by underlying physical processes (see Figure 2.8).

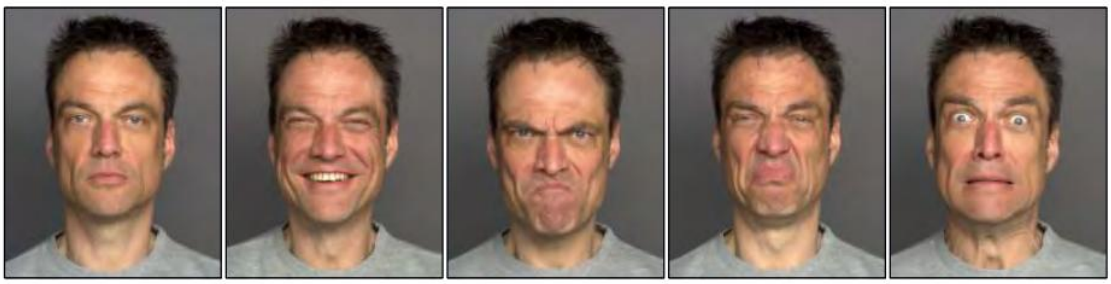

Figure 1.8 Variation of the human face. The human face contains roughly 42 muscles, so it's possible to describe most of the variation in images of the same person in the same lighting with just 42 numbers. In general, datasets of images, music, and text can be described by a relatively small number of underlying variables although it is typically more difficult to tie these to particular physical mechanisms.

This observation motivates representing each data instance using a small set of underlying latent variables. The objective is to learn the parameters that define the latent space and the mapping from this space to the observed data. In practice, the latent variables are assumed to follow a simple, predefined probability distribution. By sampling from this distribution and propagating the samples through the learned model, new data instances can be generated (Figure 1.9).

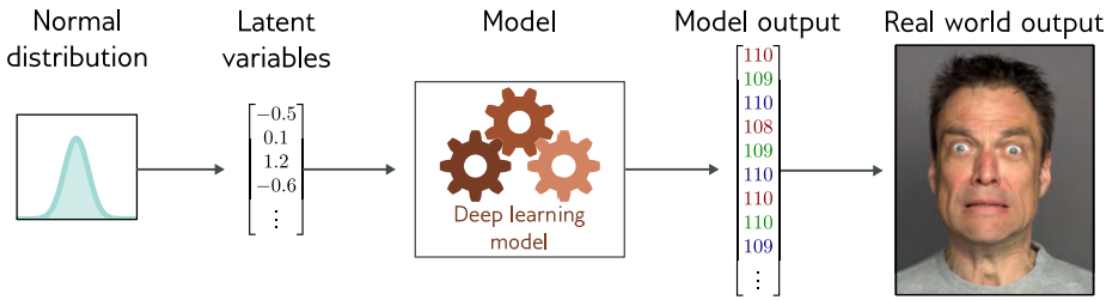

Figure 1.9 Latent variables. Many generative models use a deep learning model to describe the relationship between a low-dimensional "latent" variable and the observed high-dimensional data. The latent variables have a simple probability distribution by design. Hence, new examples can be generated by sampling from the simple distribution over the latent variables and then using the deep learning model to map the sample to the observed data space.

Such models also provide powerful tools for manipulating real data. For example, given two real data samples, one can infer their corresponding latent representations and interpolate between them. Mapping these intermediate latent points back into the data space produces smooth transitions between the original examples (Figure 1.10).

Figure 1.10 Image interpolation. In each row the left and right images are real and the three images in between represent a sequence of interpolations created by a generative model. The generative models that underpin these interpolations have learned that all images can be created by a set of underlying latent variables. By finding these variables for the two real images, interpolating their values, and then using these intermediate variables to create new images, we can generate intermediate results that are both visually plausible and mix the characteristics of the two original images.

1.2.3. Connecting supervised and unsupervised learning

Generative models based on latent variables can also enhance supervised learning tasks in which the outputs possess internal structure (Figure 2.3). For instance, consider the problem of generating an image from a textual caption. Instead of learning a direct mapping from text to pixels, one can model the relationship between latent variables that represent the text and latent variables that represent the

image. This approach offers several advantages. First, because both the input and output are expressed in lower-dimensional latent spaces, fewer paired text–image examples may be required to learn the mapping. Second, the generated images are more likely to be visually plausible, since reasonable values in the latent space tend to correspond to valid data samples. Third, by incorporating stochasticity either in the mapping between the two latent spaces or in the decoding process, the model can produce multiple distinct images that are all consistent with the same caption (Figure 1.11).

Figure 1.11 Multiple images generated from the caption "A teddy bear on a skateboard in Times Square." Generated by DALL·E-2.

1.3. Reinforcement learning

The final major area of machine learning is reinforcement learning. In this paradigm, an agent interacts with an environment and selects actions at each time step. These actions influence the state of the system, often in a stochastic manner. Executing an action may also yield a reward, and the objective of reinforcement learning is for the agent to learn a policy (a strategy for choosing actions) that maximizes expected cumulative reward. A key challenge arises from the fact that rewards may be delayed, making it difficult to determine which actions were responsible for observed outcomes. This difficulty is known as the temporal credit assignment problem. Additionally, as learning progresses, the agent must balance exploitation of strategies that are known to produce rewards with exploration of alternative actions that may lead to improved performance.

A robot is learning to play chess. The agent has a set of valid actions (chess moves) at each time step, but the outcomes are non-deterministic because the opponent can respond in many different ways. Rewards can be structured in various ways, such as assigning points for capturing pieces or providing a single reward at the end of the game for winning. Deep learning can be integrated into reinforcement learning through policy networks**,** which map observed states to actions. In the robot example, a policy network maps sensor readings to joint movements; in chess, it maps the current board state to a move choice (Figure 1.12).

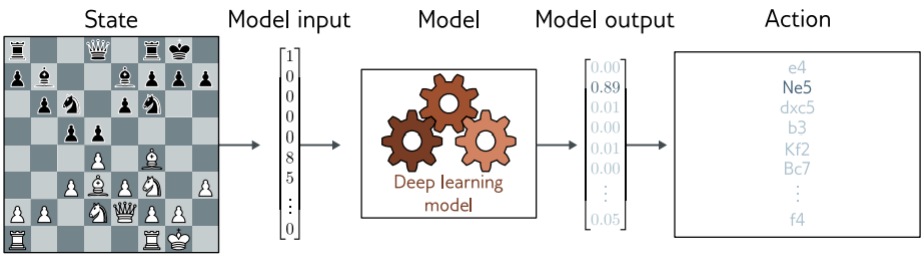

Figure 1.12 Policy networks for reinforcement learning. One way to incorporate deep neural networks into reinforcement learning is to use them to define a mapping from the state (here position on chessboard) to the actions (possible moves). This mapping is known as a policy.